

SIMATS ENGINEERING
ASSESSMENT TOOL 3

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COURSE NAME: Computer Networks

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

1. Analyze how network congestion affects packet delay and packet loss.

Network congestion occurs when too much traffic is sent through the network. Packets wait longer in router queues, increasing delay. If the buffer becomes full, packets are dropped, causing packet loss.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

Answer: Star topology is easy to manage, but if the central switch fails, the connected devices lose communication. Mesh topology has multiple paths, so communication can continue even if one link fails. Therefore, mesh provides better fault tolerance but costs more.

3. Why does increasing the number of routers between source and destination increase end-to-end delay?

Answer: Every router adds processing, queuing, transmission, and sometimes routing delays. When more routers are involved, these delays accumulate, increasing the total end-to-end delay.

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

Answer: QoS gives priority to important real-time traffic such as voice and video. It helps reduce delay, jitter, and packet loss, resulting in smoother calls and better video quality.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

Fiber optic cables provide higher bandwidth, longer transmission distances, and lower signal loss. They are also resistant to electromagnetic interference. However, fiber can be more expensive to install than copper.

6. Analyze how network segmentation using VLANs improves performance and security.

VLANs divide a network into separate logical groups. This reduces broadcast traffic and improves network performance. It also isolates departments and users, improving security and controlling access between groups.

7. Compare the effects of TCP and UDP on network performance for different applications.

TCP provides reliable delivery using acknowledgements and retransmissions, but this creates more overhead. UDP has lower overhead and is faster but does not guarantee delivery. TCP is suitable for file transfer and web applications, while UDP is useful for gaming, VoIP, and streaming.

8. Why is load balancing important in large enterprise networks?

Load balancing distributes traffic across multiple servers or network paths. It prevents individual resources from becoming overloaded and improves performance, reliability, and availability.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

Answer: Routing updates help routers learn about network changes. However, very frequent updates consume bandwidth and router processing resources. Therefore, routing updates should be timely but not unnecessarily frequent.

10. Evaluate the trade-off between network security measures and overall network performance.

Answer: Firewalls, encryption, and security monitoring improve network protection but require processing resources. This can slightly increase delay and reduce throughput. A balance should be maintained between strong security and acceptable performance.

11. A network has high delay—how will you identify and fix the issue?

Answer: I would use **ping** to measure delay and **tracert** to identify where the delay occurs. Then I would check congestion, router performance, bandwidth usage, and interface errors. I would optimize routing, reduce congestion, or increase bandwidth if required.

12. If throughput is low despite high bandwidth, what could be the reasons?

Answer: Low throughput can be caused by network congestion, packet loss, high latency, protocol overhead, faulty hardware, poor-quality cables, or limitations at the sender or receiver.

13. How would you reduce packet loss in a congested network?

Answer: I would identify the congested link, reduce unnecessary traffic, apply QoS, optimize routing, and increase bandwidth if required. I would also check for faulty cables and network interfaces.

14. A VoIP call is breaking frequently—how would you troubleshoot?

Answer: I would check packet loss, delay, jitter, and available bandwidth. I would also check router and switch interfaces and use QoS to prioritize VoIP traffic. The network path and configuration should also be verified.

15. Given a network with high jitter, suggest improvements.

Answer: I would identify network congestion, apply QoS, prioritize real-time traffic, reduce unnecessary traffic, increase bandwidth, and optimize routing. These steps can reduce variations in packet arrival time.

16. How will you simulate a congested network in Packet Tracer?

Answer: I would create a network with multiple devices and generate heavy traffic between them. I can use limited-bandwidth links and observe packet behavior in **Simulation Mode**. I would monitor packet delay, transmission, and packet drops.

17. If packets are delayed in a router, what steps will you take?

Answer: I would check the router's CPU and memory utilization, interface status, bandwidth usage, queueing, and routing table. I would identify congestion and then optimize the route, increase capacity, or apply QoS.

18. How would you optimize a network for online gaming?

Answer: Gaming requires low latency, low jitter, and low packet loss. I would use a stable connection, reduce network congestion, prioritize gaming traffic using QoS, choose efficient routes, and minimize unnecessary background traffic.

19. Design a small network and explain how you will measure its performance.

Answer: I would connect PCs to a switch and connect the switch to a router. After configuring IP addresses, I would test the network using **ping**. I would measure delay and RTT using ping, packet loss from failed packets, and throughput from successful data transfer.

20. If RTT is very high, how will you diagnose the issue?

Answer: I would first use ping to confirm the high RTT. Then I would use traceroute to identify the problematic hop. I would check congestion, routing paths, bandwidth, and router performance to locate and fix the problem.

21. Explain network performance to a non-technical person.

Answer: Network performance means how quickly and reliably information travels from one device to another. It is similar to a road: if the road is clear, vehicles travel quickly, but if there is traffic, they become slow and may not reach their destination properly.

22. Describe throughput using a real-life example.

Answer: Throughput is the actual amount of data successfully transferred in a given time. For example, if a network can theoretically transfer 100 MB per second but actually transfers 80 MB per second, its throughput is 80 MB/s.

23. Present how packet loss affects video streaming.

Answer: Packet loss means some video data does not reach the destination. This can cause buffering, freezing, poor video quality, or missing frames during video streaming.

24. Explain delay types clearly in under 1 minute.

Answer: There are four main types of network delay. Processing delay is the time required to process a packet. Queuing delay is the time a packet waits in a queue. Transmission delay is the time needed to place the packet onto the link. Propagation delay is the time taken for the signal to travel through the medium.

25. Describe Cisco Packet Tracer in a professional way.

Answer: Cisco Packet Tracer is a network simulation tool used to design, configure, test, and troubleshoot virtual computer networks. It provides simulated routers, switches, PCs, and networking protocols, allowing students and professionals to practice network configurations without physical equipment.

26. Explain QoS in simple and technical terms.

Answer: In simple terms, QoS means giving priority to important traffic. Technically, Quality of Service is a set of techniques used to classify and prioritize network traffic to control delay, jitter, packet loss, and bandwidth allocation.

27. How would you present your simulation results to a client?

Answer: I would present the network topology, testing method, measured metrics, results, problems identified, and recommended solutions. I would use simple tables or graphs to show throughput, delay, packet loss, and jitter.

28. Explain jitter using a real-world analogy.

Answer: Jitter is the variation in packet arrival time. For example, suppose a bus is expected every 10 minutes. If it arrives after 5 minutes, then 15 minutes, and then 8 minutes, the arrival time is inconsistent. This is similar to network jitter.

29. Describe how you tested network performance in a lab.

Answer: I created a topology in Cisco Packet Tracer, configured IP addresses and routing, and tested connectivity using ping. I measured delay, RTT, and packet delivery and observed the network in Simulation Mode to understand packet behavior.

30. Explain the importance of performance metrics in one structured answer.

Answer: Performance metrics help us determine whether a network is operating efficiently. Throughput measures actual data transfer, delay measures communication time, packet loss measures failed packet delivery, and jitter measures variation in packet arrival. These metrics help identify problems and improve network performance.

31. Introduce yourself and explain your knowledge of network performance.

Answer: I am a second-year B.Tech Information Technology student. I have basic knowledge of computer networking and practical experience with Cisco Packet Tracer. I understand concepts such as IP addressing, routing, throughput, delay, packet loss, RTT, and jitter. I am interested in improving my networking and troubleshooting skills.

32. Why are you interested in networking and simulation tools?

Answer: I am interested in networking because it helps me understand how different devices communicate. Simulation tools like Cisco Packet Tracer allow me to practice network configuration and troubleshooting without needing physical equipment.

33. What makes you confident in analyzing network performance?

Answer: I understand the important performance metrics and know how to apply them in practical network scenarios. I can use tools such as ping and Cisco Packet Tracer to identify issues related to delay, packet loss, and connectivity.

34. Explain a concept you are very strong in related to networking.

Answer: I am comfortable with network performance metrics. Throughput represents the actual successful data transfer, delay represents the time taken for communication, packet loss represents packets that fail to reach the destination, and jitter represents variation in packet arrival time.

35. How would you handle a question you don't know?

Answer: I would remain calm and honestly say that I do not know the complete answer. I would explain the related concepts that I understand and then learn the correct answer. I believe it is better to be honest than to provide incorrect information.

36. Describe a situation where you solved a technical problem.

Answer: While practicing Cisco Packet Tracer, I faced a connectivity problem between devices. I checked the IP address, subnet mask, default gateway, and cable connections step by step. After identifying the configuration error, I corrected it and used the ping command to confirm connectivity.

37. Why should we select you for a networking role?

Answer: I have a strong interest in networking and a willingness to learn. I have basic practical knowledge of Cisco Packet Tracer and networking concepts. I am a quick learner, patient while troubleshooting, and interested in developing my technical skills.

38. What challenges did you face while learning Packet Tracer?

Answer: Initially, configuring routers and switches, understanding IP addressing, and troubleshooting connectivity were challenging. I improved by practicing different network topologies and using commands such as ping to identify problems.

39. How do you improve your technical skills regularly?

Answer: I improve my skills through regular practice, online learning, coding exercises, and practical projects. For networking, I practice different topologies in Cisco Packet Tracer and troubleshoot connectivity and configuration problems.

40. Demonstrate confidence while explaining a complex concept simply.

Answer: I would explain network congestion using a traffic example. Network congestion is like a traffic jam. When too many vehicles use the same road, they move slowly and some may not reach their destination. Similarly, when too much data enters a network, packets experience higher delay and some may be dropped. We can reduce congestion using QoS, better routing, traffic management, or additional bandwidth.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions